CPU

File: cpu.kicad_sch

DDR3L

File: ddr3l.kicad_sch

Alimentazione

File: alimentazione.kicad_sch

Bus MIKILAB

File: bus-mikilab.kicad_sch

Storage

File: storage.kicad_sch

USB

File: usb.kicad_sch

Rete

File: rete.kicad_sch

Radio

File: radio.kicad_sch

GNSS

File: gnss.kicad_sch

Sheet: /
File: SchedaIMX6ULL.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/10

U1 NXP MCIMX6Y2CVM08AB (LMX 6ULL full feature, 792 MHz, −40..105 C. MAPBGA-289 14x14 mm passo 0.8). Stevo: IMX6ULLCEC Rev 1.2 tabelle 90/91, confrontate con la netlist di una scheda KiCad aperta: 289/289.

Boot: BOOT_MODE[1:0]=10 (boot interno) e BOOT_CFG tutti bassi con i pull-down interni da 100k (tabella 78) = QSPI, come SW501 tutto OFF sulla EVK NXP. Pull-up 2N0 su LCD_DATA04 e LCD_DATA06 (micro50). Se QSPI e' vuota il ROM va in download seriale su USB_OTG1.

Assegnazione: ECSPI2=BUS_POWER, ECSPI1=MAIN, I2C2=bus MIKROLAB, I2C1=EEPROM MAC, UART1=console, UART3=GNSS, UART8=BT, GPT1 CAPTURE1=PPS (UART2_TX); CAPTURE2=EXTINT (UART2_RX). Elenco completo in docs/pinmap-imx6ull.md.

U1A
MCIMX6Y2CVM08AB
SOC1/5

U1B
MCIMX6Y2CVM08AB
Alimentazioni

U1C
MCIMX6Y2CVM08AB
Masse

U1D
MCIMX6Y2CVM08AB
I2C1 / BOOT_CFG

U1E
MCIMX6Y2CVM08AB
NAND QSPI, uSDHC2, GPIO1 e SDS

U1F
MCIMX6Y2CVM08AB
I2C ECSPI1/2, I2C2 e ENET2

U1G
MCIMX6Y2CVM08AB
ENET1, UART, GPIO1, JTAG

U1H
MCIMX6Y2CVM08AB
USB, clock, SNVS, sistema

VDD_ARM_CAP / VDD_SOC_CAP: uscite LDO interni, solo condensatori

VDD_HIGH_CAP (a NVCC_DRAM_2P5), NVCC_PLL, VDD_SNVS_CAP, VDD_USB_CAP

VDD_SOC_IN 1.4 V

VDD_HIGH_IN e NVCC 3.3 V: un 100 nF per dominio (CSI, ENET, GPIO, LCD, NAND, SDI, UART)

NVCC_DRAM 1.35 V (solo SoC)

VDD_SNVS_IN

Y1
24MHz

Carico 24 MHz (CL da verificare sul quarzo YXC)

Y2
32.768kHz

Carico RTC: circa 2xCL meno parassite (datasheet, tabella 7)

Boot: BOOT_MODE=10. rifiniti a VDD_SNVS: BOOT_CFG1[4] a [6] DNP finche' lo strap QSPI non e' confermato sulla fase map: JTAG_MOD a GND con 4.7k (tabella 5)

SW1
RESET

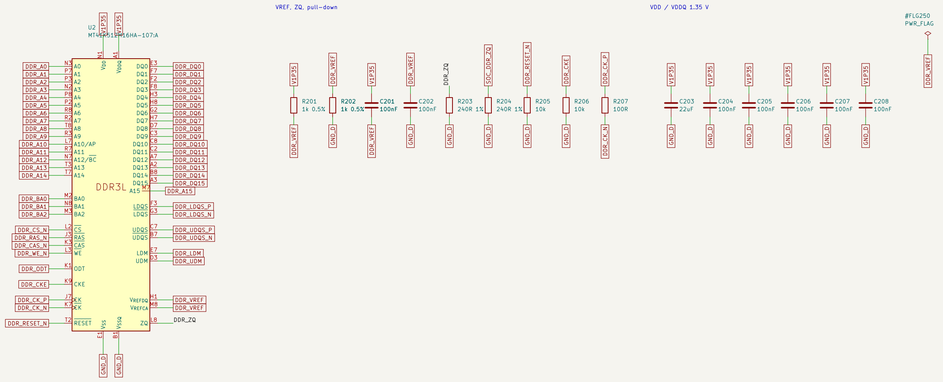
SW2
ON/OFF

J5
Console UART1

Pull-up open drain USB2_OC_N. PWR_SRC_ST

U2 Micron MT41K512M16HA-107:A. 8 Gb x16 = 1 GB. FBGA-96 9x14 passo 0.8. a 1.35 V. Un solo chip punto-punto: niente VTT. Simbolo da MT41K256M16HA di KiCad con A15 su M7 (ball map JEDEC x16).

VREF a partitore 1k 0.5% con 100 nF su ogni resistenza (datasheet 60ULL, tabella 3), unica per DRAM_VREF, VREFCA, VREFDQ. ZQ 240R 1% su SoC e DRAM. RESET# e CKE con 10k a GND. CK con 100R differenziale vicino alla DRAM (da confermare in simulazione).



VREF, ZQ, pull-down

VDD / VDDQ 1.35 V

Sheet: /DDR3L/		
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Title:		
Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/10

VIN_SYS 12 V dal BUS_POWER -> TPS56637 5 V (U22) -> TPS2121 U17 (IN1 bus con priorità, IN2 VBUS USB-C 1) -> 5V_SYS -> buck 3.3 V (U23), 1.35 V DDR (U26), 1.4 V VDD_SOC_IN (U27), 3.3 V SSD (U39).

Sequenza 3.3 -> 1.35 -> 1.4 V a RC sugli EN del TLV62569 (soglia EN max 1.2 V): 100k/470nF circa 21 ms, 100k/1uF circa 45 ms. POR_B dal TPS3839K33 (2.93 V, push-pull) con 1k in serie, così SW1 può cortocircuitare POR_B.

VDD_SNVS: OR a diodi BAT54C tra 3V3 e CR1220 (BT1), alimenta anche V_BCKP del GNSS. Nessuna ricarica della pila. Passivi 0603 minimo: 47 uF e 10 uF/25 V in 1206, 22 uF e 10 uF in 0805.

TPS56637: EN 169k/36.1k, FB 73.2k/10k = 4.99 V, FF 20k+100pF, MODE aperto = FCCM, 2x47 uF al posto di 4x22 uF

TPS2121 (come RK3568): PR1 4.49 V, OV 5.98 V, ILM 22.1k = 4–5 A, SS 100 nF

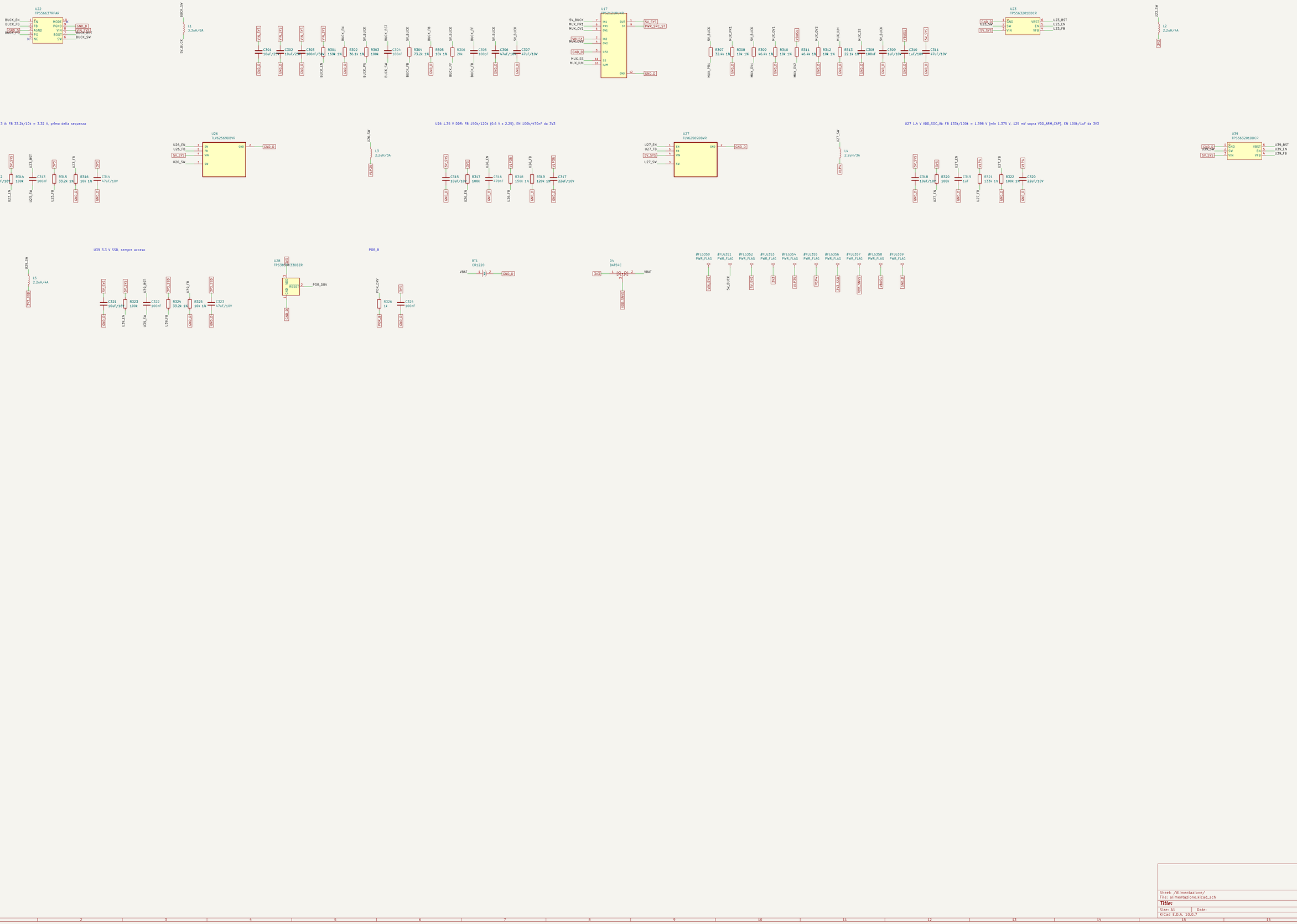
U23 3.3 V / 3 A: FB 33.2k/10k = 3.32 V, primo della sequenza

U26 1.35 V DDR: FB 150k/120k (0.6 V x 2.25), EN 100k/470nF da 3V3

U27 1.4 V VDD_SOC_IN: FB 133k/100k = 1.398 V (min 1.375 V, 125 mV sopra VDD_ARM_CAP), EN 100k/1uF da 3V3

U39 3.3 V SSD, sempre acceso

POR_B



Sheet: /Alimentazione/
File: alimentazione.kicad_sch
Title:
Size: A1 | Date: | Rev:
KiCad E.D.A. 10.0.7 | Id: 4/10

J11 BUS_POWER 30 pin (Hirose FH12-30S) su ECSPI2. J12 MAIN 28 pin REV5.1 (XUNPU FPC-05F-28PH20, pin 28 = HIT#) su ECSPI1. Mappa del connettori identica a SchedaA133/RK3568. Rail BUS_* passanti, non collegati ai rail della scheda.

Serie 33R solo sul clock SPI. Pull-up 4.7k su ALERT#/HIT#, 2.2k su I2C. VREF_ENABLE 10k e VBIAS_ENABLE 100k a GND. CS e FAULT con i pull-up interni del 6ULL (pad control nel device tree). SPI_ID a GND diretti. ESD TPD4E05U06DQA sulle linee logiche.

J11
BUS_POWER_MIKILAB

J12
BUS_MIKILAB_MAIN

Serie SPI 33R, solo clock

R401 33R
R402 33R

Livelli di riposo

R403 10k
R404 100k
R405 4.7k
R406 4.7k
R407 2.2k
R408 2.2k

U29
TPD4E05U06DQAR

U30
TPD4E05U06DQAR

U31
TPD4E05U06DQAR

U32
TPD4E05U06DQAR

U33
TPD4E05U06DQAR

U34
TPD4E05U06DQAR

Disaccoppiamento connettori

C401
100nF

#FLG450
PWR_FLAG

#FLG451
PWR_FLAG

#FLG452
PWR_FLAG

Sheet: /Bus MIKILAB/		
File: bus-mikilab.kicad_sch		
Title:		
Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 5/10

U5 W25Q128JV su QSPI1 porta A (tabella 79, boot). J5 microSD DM3AT su uSDHC1 (sul BOTTOM). 3.3 V. CD su UART1_RTS_B (USDHC1_CD_B). U25 protegge DAT0–3: CMD e CLK non hanno ESD (un solo TPD4E in BOM, 4 linee).

U38 JMS578 (QFN-48 6x6 passo 0.4): solo USB 2.0, SuperSpeed non collegati. Datasheet Rev 1.01: 12K su REXT, 10 nF serie su SATA, 4.7 uH su LXO –> 1.2 V (VCCK/AVDDL), 30 MHz. MODE[1:0]=01, TME=0. VREG_IN e VBUS da 5V_SYS. VCCO dal suo AV330.

Niente SPI NVRAM del JMS578 (facoltativa, solo VID/PID vendor e firmware aggiornabile): usa il ROM interno. J13 solo SSD SATA: DEVSLP basso, PEDET libero. Nessun reference JMicron pubblico: da confermare sul primo prototipo.

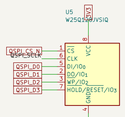
QSPI: 22R su SCLK, pull-up 10k su CS/IO2/IO3

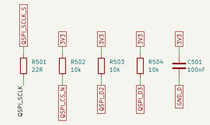
microSD: 22R su CLK, pull-up 10k su CD; CMD e DAT con i pull-up interni del 6ULL

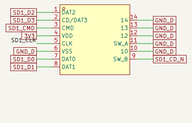
JMS578

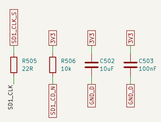
SATA: 10 nF serie (datasheet)

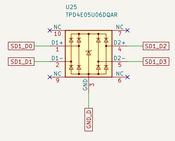
M.2

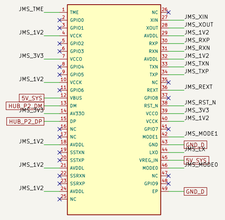


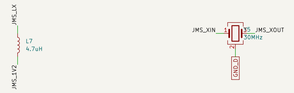






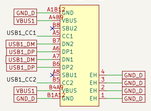


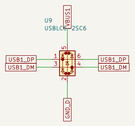




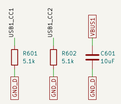
J3 USB-C 1: USB_OTG1, device/download seriale e alimentazione (VBUS1 -> IN2 del TPS2121). Sink: 5.1k su CC1/CC2. J8 USB-C 2 host: porta 1 dell'hub, VBUS dal TPS2553 (~1 A). Rp 56k. Le due USB-C sono affiancate sullo stesso bordo.

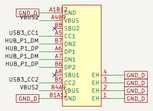
U36 FE1.1s Rev B su USB_OTG2 (3.3 V esterni su VD33F, VDD5 e pin 12/13/28 liberi come da datasheet Rev B12.4): porta 1 J8, porta 2 JMS578, porte 3–4 libere, VBUSM e BUS1 a 3.3 V (host sempre presente, self-powered). USB_OTG2_VBUS del 6ULL da 5V_SYS.

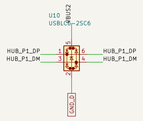




CC sink J3

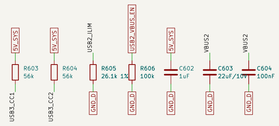


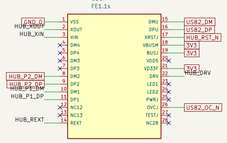






J8 host: Rp 56k. RILIM 26.1k. EN 100k a GND

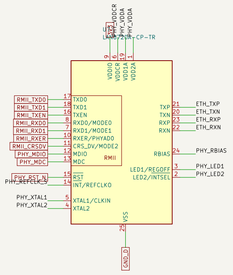


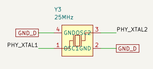


Hub: REXT 2.7k, XRST/ 10k, DRV basso = hub normale

U7 LAN8720A su ENET1 RMII. Y3 25 MHz montato: il PHY genera i 50 MHz (nINTSEL basso) su REFCLK0 verso ENET1_TX_CLK (REF_CLK in ingresso, SION). nINT quindi non disponibile. Strap MODE=111 (10k su RXD0/RXD1/CRS_DV). PHYADD=0.

J1 HR911105A: magnetici e LED integrati. TX/RX con 49.9R a VDDA, CT a VDDA con 22 nF. Polarita' dei LED dal disegno Hanrun: da verificare (solo indicazione, non blocca il link). U6 24AA02E48 su I2C1 con il MAC EUI-48.

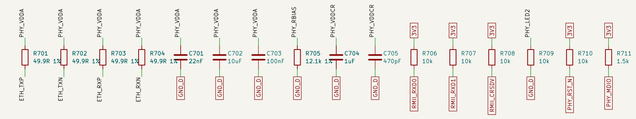


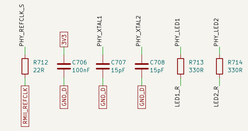




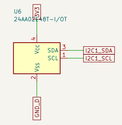


PHY: terminazioni, RBIAS 12.1k, VDDCR 1uF+470pF, straps, REFCLK 22R





EEPROM MAC: I2C1



Sheet: /Rete/
File: rete.kicad_sch

Title:

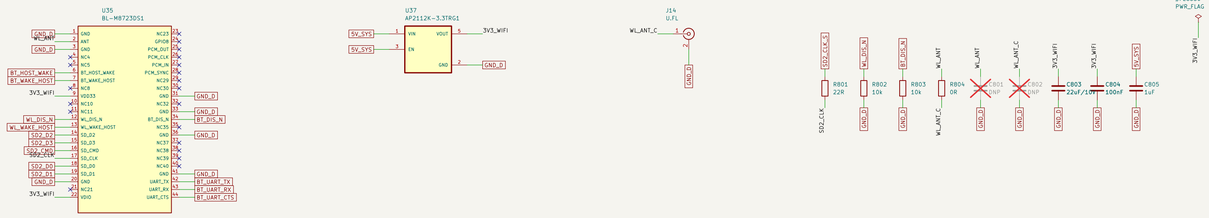
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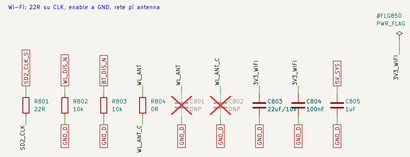
KiCad E.D.A. 10.0.7 | Id: 8/10

U35 LB-LINK BL-M8723DS1 (RTL8723DS, 12x12 mm, datasheet V1.0.1.3): Wi-Fi su uSDHC2 (pad NAND), BT su UART8 (pad ENET2). VDD33 e VDIO da U37 (3.3 V dedicati ai picchi TX). WL_DIS# e BT_DIS# senza pull-up nei moduli: 10k a GND, li alza Linux.

SDIO con i pull-up interni del 6ULL. Antenna su U.FL J14 (scelta del placement: la BOM lasciava u.FL o PCB) con rete pi OR + 2 DNP da tarare. UART_CTS del modulo da UART8_RTS. Il modulo non ha RTS (pin 41 = GND): H5 a 3 fill. PCM non usata: PCM_OUT libero (strap da non alzare).

Wi-Fi: 22R su CLK, enable a GND, rete pi antenna

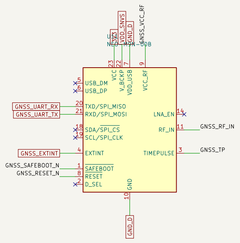




Sheet: /Radio/		
File: radio.kicad_sch		
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Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 9/10

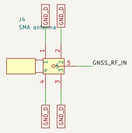
U11 u-blox NEO-M9N-00B su UART3. TIMEPULSE -> 33R -> GNSS_PPS su GPT1_CAPTURE1 (UART2_TX) e buffer U20 -> 33R -> header J7. EXTINT IN da J7 -> ESD + 100R -> buffer U19 -> GNSS_EXTINT su NEO e GPT1_CAPTURE2 (UART2_RX): evento marcato da entrambi.

J4 SMA antenna, bias 3.3 V attiva: VCC_RF -> 10R -> 27 nH (0603) -> RF_IN, 10 nF. Linea 50 ohm corta. V_BCKP da VDD_SNVS (CR1220 via BAT54C). J7: PPS OUT. EXTINT IN. GND (come scheda Tibbo).



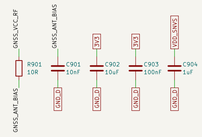




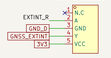




Bias antenna e alimentazione











Timing: serie e buffer